Measures of Dispersion

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Dispersion

- Measures of dispersion measure how spread out a set of data is
- how much variability there has in the data
- Two data with same mean, can have different variability

Dispersion

- For example, the monthly income (in $1000) of 5 randomly selected people is,
- Sample A: 60, 62, 60, 58
- Sample B: 102, 80, 10, 48

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- Sample B: 102, 80, 10, 48 $(\bar{X}_B = 60)$

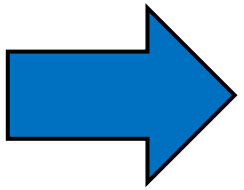
Dispersion

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- Sample A: 60, 62, 60, 58 $(\bar{X}_A = 60)$

- Sample B: 102, 80, 10, 48 $(\bar{X}_B = 60)$

So it is also important to describe "How much the observation vary from one to another



Dispersion/ Variation

Why do we need?

- To determine the reliability of an average
- To compare the variability of two or more data sets.

Types of Measures of Dispersion

- Two types of measures of dispersion

1. Absolute measure

2. Relative measure

Absolute measures

- Only measure the inherent variation of a data set

- Important absolute measures are-

1. Range
2. Mean Deviation
3. Variance
4. Quartile Deviation

Range

- $Range = Highest\ value - Lowest\ value$

- For example, age of 5 randomly selected people (in years)

$$24, 18, 36, 20, and\ 22$$

Range

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$$(Range = ????)$$

Range

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- For example, age of 5 randomly selected people (in years)

$$24, 17, 36, 20, and\ 22$$

$$Range = 36 - 17 = 19\ years$$

Mean Deviation

- Let $x_1, x_2, \ldots, x_n$ be n observations, where $\bar{x}$ is average/ or mean
- $MD = \frac{\sum |x_i - \bar{x}|}{n}$ [for ungrouped data]
- $MD = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i}$ [for grouped data]

Mean Deviation

- For example, $x = 2,3,5,6$. Calculate the mean deviation.

x_i
2
3
5
6

Mean Deviation

- For example, $x = 2,3,5,6$. Calculate the mean deviation.

x_i	$\bar{x}$	$\lvert x_i - \bar{x} \rvert$
2	$\frac{\sum x_i}{n} = 4$	2
3		1
5		1
6		2

$$MD = \frac{\sum |x_i - \bar{x}|}{n} = \frac{6}{4} = 1.5$$

Mean Deviation

Age (in years)	10-20	20-30	30-40	40-50	50-60
f_i	3	6	10	4	2

Calculate the mean deviation about mean. (Ans: 8.448)

Variance

- Let $x_1, x_2, \ldots, x_n$ be n observations

For grouped, $\sigma^2 = \frac{\sum f_i(x_i-\mu)^2}{N}$

- $\sigma^2 = \frac{\sum(x_i-\mu)^2}{N}$ [Population variance (ungrouped)]

$$\sigma^2 = \frac{\sum x_i^2 - N\mu^2}{N}$$

- $s^2 = \frac{\sum(x_i-\bar{x})^2}{n-1}$ [Sample variance (ungrouped)]

$$s^2 = \frac{\sum x_i^2 - n\bar{x}^2}{n-1}$$

For grouped, $s^2 = \frac{\sum f_i(x_i-\bar{x})^2}{n-1}$

Variance

- For example, the salaries of randomly selected 10 newly graduates are,

25,32,41,27,35,42,29,37,45,35

Calculate the variance

Variance

x_i	$\bar{x}$	$(x_i - \bar{x})^2$
25	34.80	96.04
32		7.84
41		38.44
27		60.84
35		0.04
42		51.84
29		33.64
37		4.84
45		104.04
35		0.04

Sum=397.9

Variance

- For example, the salaries of randomly selected 10 newly graduates are,

25,32,41,27,35,42,29,37,45,35

Variance, $s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1} = \frac{397.9}{9} = 44.21$

Standard Deviation

- Square root of variance
- $\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$ [Population standard deviation (ungrouped)]
- $s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$ [Sample variance (ungrouped)]

Standard Deviation

- For example, the salaries of randomly selected 10 newly graduates are,

25,32,41,27,35,42,29,37,45,35

a) Calculate the standard deviation

b) Interpretation of your result

Standard Deviation

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Standard Deviation

- For example, the salaries of randomly selected 10 newly graduates are,

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a) Calculate the standard deviation

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}} = \sqrt{44.21} = 6.65$$

b) Interpretation of your result

The average variation of the salaries of 10 randomly selected newly graduates from the mean salaries is 6.65

Relative measures

- To compare the variation of two or more data sets having different or same units.

1. Coefficient of Range
2. Coefficient of MD
3. Coefficient of variance
4. Coefficient of QD

Coefficient of variance (CV)

- It is a ratio
- Between standard deviation (SD) and mean
- Expressed as percent

$$\text{General formula of CV} = \frac{SD}{mean} \times 100$$

Coefficient of variance (CV)

- $CV = \frac{\sigma}{\mu} \times 100$; $[For\ population]$
- $CV = \frac{s}{\bar{x}} \times 100$; $[For\ sample]$



Higher CV =
High variation

Lower CV =
Low Variation

More consistent
More uniform
More homogeneous
More stable

Coefficient of variance (CV)

- Sample A: Mean = 32 and SD = 4

$$CV_A = \frac{4}{32} \times 100 = 12.5\%$$

- Sample B: Mean = 25 and SD = 5

$$CV_B = \frac{5}{25} \times 100 = 25\%$$

- Which section is more consistent in getting final exam mark?

Since, the CV of A is lower than the CV of B.
Thus, sample A is more consistent comparatively sample B

Thank You !